

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS
Doctoral Program in Deep Tech with a Focus on Artificial Intelligence
and Emerging Technologies

DELIVERABLE No. 3 – RESEARCH PROTOCOL

01 TITLE

AI Governance Maturity Model for Peruvian Organizations within the Framework of Law No. 31814 and NTP-ISO/IEC 42001:2025: Design, Validation, and Automated Compliance Assessment

02 ABSTRACT

This study proposes the design and validation of an AI Governance Maturity Model for Peruvian organizations. A systematic literature review guided by PRISMA 2020 will identify governance dimensions, maturity levels, indicators, and assessment methods. The evidence will be aligned with the Peruvian regulatory context and ISO/IEC 42001. The study will also explore LLM-supported documentary assessment. The expected contribution is a practical framework for assessing and progressively improving organizational AI governance.

03 INTRODUCTION & PROBLEM STATEMENT

AI adoption creates challenges involving accountability, transparency, risk, privacy, security, ethics, human oversight, and regulatory compliance. Although international standards and Peruvian AI regulation provide guidance, organizations still require a contextualized mechanism to measure governance maturity. The problem addressed is the absence of a validated model that enables Peruvian organizations to assess current capabilities, identify gaps, and define improvement priorities.

04 LITERATURE REVIEW

The review will examine AI governance, Responsible AI, AI management systems, maturity models, risk management, governance assessment, and organizational AI capabilities. Searches will cover Scopus, Web of Science, IEEE Xplore, ScienceDirect, SpringerLink, and ACM Digital Library, complemented by relevant standards, regulations, and institutional frameworks. PRISMA 2020 will guide transparent reporting of study identification, screening, eligibility, and inclusion.

05 QUESTIONS, HYPOTHESIS & OBJECTIVES

RQ1: What dimensions, maturity levels, indicators, and assessment methods should constitute an AI Governance Maturity Model suitable for Peruvian organizations? RQ2: How can LLMs support documentary assessment against selected governance requirements? Hypothesis: integrating evidence-based international practices with Peru's regulatory context and ISO/IEC 42001 can produce a valid and measurable maturity framework. Objective: design and validate the model and evaluate the feasibility of LLM-supported assessment.

06 METHODOLOGY

A Systematic Literature Review will use combinations of "Artificial Intelligence/AI", "Governance/Responsible AI", and "Maturity Model/Assessment/Governance Framework". English/Spanish relevant studies and authoritative frameworks will be included; duplicates, insufficiently documented studies, and purely technical studies without governance implications will be excluded. Records will be deduplicated, screened, assessed in full text, and reported using a PRISMA 2020 flow diagram. Data will be extracted and synthesized thematically, mapped to Peru and ISO/IEC 42001, followed by model design, expert validation, empirical application, and evaluation of LLM-assisted documentary assessment.

07 ETHICAL CONSIDERATIONS

The literature review uses published documentary sources. Academic integrity will be ensured through transparent procedures, accurate citation, traceable extraction, and faithful reporting. If experts or organizational participants are involved in validation, informed consent, confidentiality, voluntary participation, data protection, and institutional ethical requirements will apply. LLM use will consider privacy, security, bias, transparency, traceability, and human oversight.

08 EXPECTED RESULTS

Expected outputs are a validated AI Governance Maturity Model with defined governance dimensions, maturity levels, measurable indicators, and assessment criteria for Peruvian organizations; identification of international approaches and research gaps; regulatory and standards mapping; and evidence on the potential and limitations of LLM-supported documentary assessment. The model should support diagnosis, gap identification, prioritization, and continuous improvement.

09 TIMELINE & BUDGET

Indicative 24-month plan: SLR and theoretical framework (M1-6); regulatory mapping and model design (M7-10); expert validation (M11-14); empirical application (M15-18); LLM assessment evaluation (M19-21); analysis and dissertation writing (M22-24). Budget categories: researcher time, standards/database access, analysis software, LLM/computing resources, validation logistics, and dissemination. Exact costs will depend on institutional access and study scope.

10 BIBLIOGRAPHY

Page, M. J., et al. (2021). The PRISMA 2020 statement. *BMJ*, 372, n71. Shamseer, L., et al. (2015). PRISMA-P 2015: elaboration and explanation. *BMJ*, 349, g7647. ISO/IEC 42001:2023, Artificial intelligence - Management system. Congreso de la Republica del Peru. (2023). Law No. 31814, promoting the use of artificial intelligence for the country's economic and social development.